

RL Based UAV Trajectory Optimization to Avoid Jammers and Optimize Wireless Connectivity

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Abstract—Drones have found extensive use in military applications owing to their inconspicuousness and easy deployment. Major challenges in the UAV space today include: (a) Maintaining seamless connectivity with the ground station to maintain command and control (C2) link and ensure continuous streaming of high quality data and (b) avoiding enroute RF jammers that can saturate the RF receiver and inhibit communication. Additionally, GNSS jammers can render drones completely non-functional by not allowing the drone to communicate with GPS satellites to ascertain their location. This work addresses these two challenges by developing a reinforcement learning (RL) algorithm that is able to optimize the wireless connectivity while avoiding jammers along the drone route. Notably, the algorithm ensures minimal deviation from the planned path in the presence of jammers. The training process for the drone can be completely accomplished pre-flight. The drone is only required to carry a lookup table (Q-table) during flight thereby reducing the computing requirements onboard the drone.

Index Terms—drone, UAV, wireless, path planning, reinforcement learning, BVLOS, anti-jamming

I. INTRODUCTION

Drones have become a ubiquitous tool in modern warfare, with both state and non-state actors utilizing them for a variety of tasks. The US military, in particular, has heavily invested in drone technology, deploying them for a variety of operations that include Intelligence, Surveillance, and Reconnaissance (ISR), targeted strikes, logistics and resupply, and search and rescue. Planning the path for such sensitive drone operations is not straight-forward, since employing just conventional path planning objectives, such as distance traveled or minimization of flight time may not always be suitable. For instance, maintaining a good quality C2 link over the drone route is critical for sensitive missions, and may take priority over the shortest distance. Additionally, if a drone is deployed for surveillance, it is important that the drone trajectory always lies in regions of good coverage so that data can be continuously streamed. Concurrently, drones employed for defense purposes are always in threat of jamming, which involves shooting a focused high power antenna beam at the drone with the intention of saturating the front-end of the drone modem receiver rendering it impossible for the drone to communicate with the ground station (or GPS satellites for GNSS jamming) thereby disrupting service. There is thus a need to simultaneously optimize the drone path both in terms of signal connectivity and avoiding jammers whose locations might not be known apriori. The first challenge

to accomplish such path planning is the dynamicity of the wireless channel. A wireless channel can be impacted by a variety of factors such as geography, weather, user traffic, etc., making it impossible to build a standard channel model that can be used to compute the optimal route a priori. This challenge is further compounded by the presence of jammers whose locations might be unknown. This paper proposes a reinforcement learning approach that allows the drone to always fly in regions of good connectivity while avoiding multiple jammers on its route. Further, it is shown that the algorithm only deviates minimally from a pre-planned path in the vicinity of a jammer, which could potentially be an important concern for sensitive missions.

Classical anti-jamming techniques include using adaptive filters that can attenuate the jammer signal while boosting the signal to be transmitted, or Frequency Hopping Spread Spectrum (FHSS) where the frequency of the required information is dynamically varied to avoid its corruption from a jammer [1]. However, such hardware implementations might not conform to the size, weight, power and cost (SWAP-C) constraints posed by the drone form factor. There have been limited publications that discuss the issue of trajectory optimization to avoid jammers. The authors in [2] propose successive convex approximation to optimize the path of the drone in 3D space to avoid jammers. In [3], the downlink power is constantly monitored, and the jammer location is estimated by attempting to find a region over which the interference levels are within a certain threshold. However, the requirement for the drone to still be in regions of good connectivity while the jammer is being avoided is not considered in these works. It is worth noting that there have been several publications that address the idea of combating UAVs acting as jammers for ground terminals. For instance, the authors in [4] and [5] address the issue of UAVs acting as jammers for ground terminals and vehicle ad-hoc networks respectively. On the other hand, several publications have addressed optimizing the drone path trajectory to optimize communication parameters. For instance the authors in [6], [7] and [8] use reinforcement learning for path planning with an objective of flight time and communication outage duration. In [9], the authors use reinforcement learning to optimize a joint objective of UAV upload throughput maximization and minimum interference to ground mobile devices. In contrast, our algorithm enables

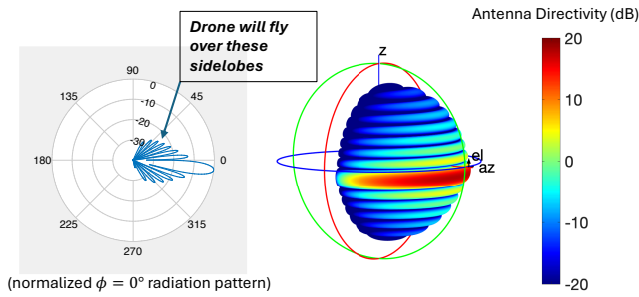


Fig. 1. Representative base station antenna beam, with directivity of 18 dBi with a down-tilt of 7°.

the drones to learn key characteristics of the wireless channel enabling it to always fly in regions of good coverage, while avoiding jammers. Moreover, our algorithm requires no apriori knowledge of the location of the jammers making it very suitable for highly sensitive operations. Finally, our algorithm is developed with the intent of accomplishing the training when the drone is on the ground. While flying, the drone simply carries a ‘look-up’ table that enables it to take optimal action at any given state. This significantly reduces the computing requirements, thereby meeting the SWAP-C requirements posed by drones.

II. WIRELESS LINK MODELING

In order to generate data for training and evaluating our RL algorithm, it is essential to develop a representative RF simulator that can capture the fundamental trends in the signal power at the altitude that the drone flies. For the purposes of this paper, we assume that the drone is flying in a cellular environment and thus connects to the wireless network via the cellular base stations. However, the methodology presented in this paper can be generalized to emulate any other form of wireless connectivity viz. WiFi, satellite, etc.

A. Wireless Heatmap Modeling

We begin with the assumption that the link between the drone and base station will be close to a free space link. Thus, we model the link between the drone and base station through the well known Friis equation: $P_r = P_t G_t G_r (\lambda / (4\pi R))^2$, where P_r is the received power at the drone, P_t is the base station transmit power, G_t and G_r are the transmit and receive antenna gains, λ is the wavelength of the transmitted signal’s carrier, and R is the distance between the drone and the base station. Note that G_t and G_r are in general spatially dependent. The Friis equation provides us with the downlink receive power.

Since the drone antenna would be almost omnidirectional, we assume that the gain of the drone antenna is constant over the solid angle subtended by the base station antenna over the drone.

For the base station antenna, a 15 element linear array, with an inter-element spacing of 0.5λ , operating at 2.0 GHz is assumed. This array provides a peak directivity of 18 dBi.

The array is phased to provide a down-tilt of 7 deg [10] (see Fig. 1). The individual elements are assumed to be dipole-like with a 45 deg beamwidth. Consequently, a drone that is flying in the sky will never see the main beam; it will only see the sidelobes of the base station antenna (which is at least 13 dB lower than the main beam). This makes the drone extremely sensitive to any high power signals (from jammers) that can shadow the cellular signals.

For each drone position, we evaluate the power received from each sector of each individual base station. This maximum power is now the received signal, and the powers from all other base stations (each with 3 sectors) as well as the remaining two sectors of the serving base station are treated as interference. Note here that the assumption is that all the base stations (and the three sectors corresponding to each) radiate at the same frequency. Additionally, the receiver noise is given by $P_N = \kappa T B$, where P_N is the noise power, κ is the Boltzmann constant, T is the receiver noise temperature in Kelvins, and B is the bandwidth in Hz of the communication signal [11]. Dividing the maximum power by the interference and noise power, thus calculated, constitutes the baseline Signal to Interference and Noise Ratio (SINR).

B. Jammer Modeling

To model a jammer, we assume a highly directive antenna pointed directly upwards towards the drone. Specifically, we assume a planar array consisting of 10×10 array of elements spaced 0.5λ apart to generate a main beam with a directivity of 25 dBi with a beamwidth of 5° in both the principle planes. The far-field patterns are shown in Fig. 2. We assume here that the jammer radiates the same power and frequency as that of the base stations. A key differentiator between the base station and the jammer is that the main beam of the jammer is pointed directly at the sky. This would make the power density radiated by the jammer towards the drone several orders of magnitude higher than the power density radiated by the base station antenna, which would only communicate with the drone via sidelobes (See Fig. 1). The power thus received from the jammer is considered as an additional interferer in the SINR computation described in the previous section.

It should be noted here that we have only chosen representative antenna patterns and the radiated power values (for jammers and the base stations) in order to evaluate our RL algorithm. In reality, these will need to be tailored for the specific environment that the drone will be deployed in. Nonetheless, the methodology to compute the SINR as well as the RL algorithm are independent of these parameters and can be used in any environment.

III. REINFORCEMENT LEARNING MODEL

Our algorithm utilizes Q-based learning, which is a model-free RL technique [12] [13]. The fact that this algorithm does not require any apriori knowledge about the environment makes it very suitable for a wireless environment, which is highly dynamic and unpredictable. During training, an estimate Q of the optimal state-action function Q^* is generated

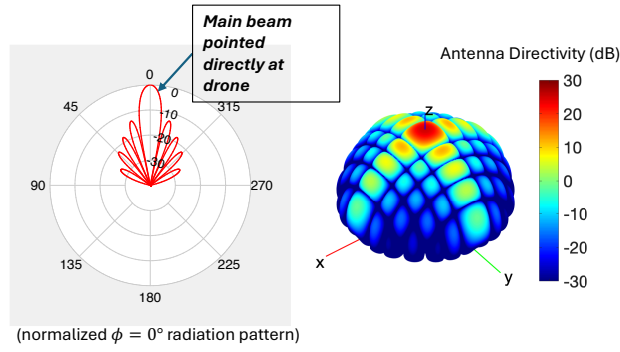


Fig. 2. Jammer antenna modeling illustrating an antenna beam with a directivity of 25 dBi that points directly towards the drone.

as the drone explores the environment. The Q-function maps state-action pairs to the value of taking action a in state s . The training process consists of an iterative update to the so-called Q-learning equation, which is derived from the Bellman optimality condition. At iteration t , if the agent is in state s_t , and takes action a , causing a change of state to s_{t+1} , the state-action estimate for (s_t, a) is updated according to:

$$Q(s_t, a) \leftarrow Q(s_t, a) + \alpha [r(s_t, a, s_{t+1}) + \quad (1)$$

$$\gamma \arg \max_{b \in A_{s_{t+1}}} Q(s_{t+1}, b) - Q(s_t, a)] \quad (2)$$

where

$r(s_t, a, s_{t+1})$ is the (immediate) reward associated with a transition from state s_t to s_{t+1} through action a ; α is a learning rate, controlling the convergence properties of the algorithm; γ is a discount factor, which discounts the future reward; and $A_{s_{t+1}}$ is the set of available actions in state s_{t+1} . After training, a policy π_Q that selects an action given a state is defined as

$$\pi_Q(s) = \arg \max_{a \in A_s} Q(s, a) \quad (3)$$

In classic Q-learning, the A_s is a fixed set A that does not change with each iteration. In contrast, our algorithm employs *constrained Q-learning* [14], which restricts the action space at any given iteration to only those actions which both will not increase the agent's distance from its destination location and respect geofencing. This leads to superior policies [14] and ensures that the drone will never fail to arrive for poorly chosen actions. It additionally guarantees that the drone will make steady progress to its destination while retaining some freedom to explore and exploit from the available actions.

During training, actions are chosen to explore the state-action space along with maximizing the reward. To achieve this, a small positive ϵ is chosen. Then, with probability $1 - \epsilon$, the action is chosen to maximize the Q function, and otherwise, the action is chosen randomly. (This is called ϵ -greedy Q-learning). Random actions should dominate the training in the initial phase of training (for exploration), whereas in the later stages, the actions with maximum Q should dominate (for convergence). To ensure this, we start the algorithm with

ϵ being almost 1. As the training episodes increase, the value of ϵ is reduced by multiplying it iteratively by a decay factor less than 1.

It is of importance to note that training is performed offboard, where, unlike the drone, computation and power requirements are less constrained. Our training was performed on an Apple MacBook Air with Apple Silicon M1 CPU. The output of the training is essentially a Q-table, which serves as a lookup table for executing the policy on the drone during flight. Thus, computational requirements for the deployed algorithm are minimal and can be easily run even on drones with modest computational capabilities.

A. State space selection

The selection of the state space, reward, and action space is the first design decision in developing the path optimizer. It might be intuitive to define the state of the drone by its position and the reward can be a wireless metric like SINR. The actions can thus correspond to brief maneuvers which take the drone from one state (position) to another. This implies that once trained, the policy essentially informs the drone what to do when it is at a certain *location*. Thus, if the SINR during evaluation is different from the one that it has seen in its training phase, it will take actions optimized for the training environment, which are not necessarily optimal for the evaluation environment.

There is thus a necessity to define the state space so that it includes quantities that are generalizable, i.e., not specific to any particular training or deployment environment. In a wireless environment, SINR is primarily dependent on the spatially-dependent transmitter antenna gain, receiver noise, and interference. By defining the state space appropriately, and then training over a large set of environments the drone can learn how to fly optimally in an unseen deployed environment. Concurrently, the presence of a jammer will significantly increase the dynamic range of the SINR in the heatmap (since the jammer will create a very deep null in the heatmap). To address this issue, we utilize the fact that every SINR can be mapped to a certain Channel Quality Indicator (CQI) that is quantized and is thus much more amenable to Q-learning [15]. For this paper, we enhance our state space to include the following four parameters: CQI in the current position, CQI in the previous position (at the previous iteration of the episode), action that it took at the last iteration, and the distance to the nearest base station. Such a state space allows the drone to learn the trends of the antenna pattern (and the associated signal quality), making the algorithm more applicable to a wide variety of environments.

IV. REPRESENTATIVE EXAMPLES

In this section we demonstrate the utility of our algorithm. The environment is assumed to be a hypothetical urban area with a cluster of nine cellular base stations placed uniformly in a grid with spacing of 5 km. Each base station consists of three sectorized antennas each serving a 120° field of regard. A hypothetical drone is now trained to fly in this environment

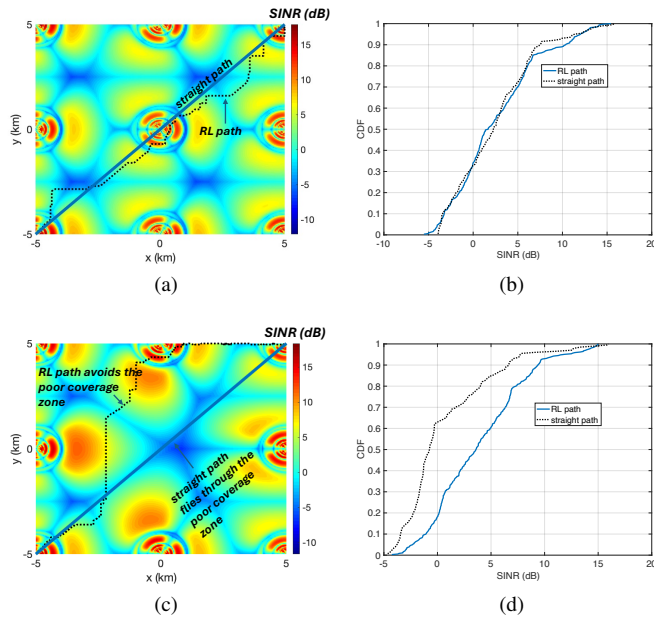


Fig. 3. Comparison of RL path (black dotted line) with the straight line path plotted against (a) a SINR heatmap. The CDF of the two paths are compared in (b). The assumption is that the training and the evaluation environment are identical. (c) and (d) represent the drone trajectory and CDF, respectively, in an environment different than the training environment where poor coverage zone is created at the center of the heatmap.

from a starting point at the bottom left corner of the grid to an end point at the upper right corner. The altitude is assumed to be 150 m and the bandwidth is assumed to be 1 MHz. For the current implementation, we use the SINR, scaled to lie between 0-1, as the reward.

We simulate 40000 episodes and use ϵ -greedy Q learning to learn a policy for dynamic path planning in this environment. Each episode involves the drone taking actions within a training environment until the goal is reached or a maximum number of steps is exceeded. Each action by the drone and the associated rewards leads to an update of the Q-table. Fig. 3a illustrates the SINR heatmap. As described in Section II-A, this map is generated from the 9 base stations, each having 3 individual sector antenna patterns. The antenna gain at each location is the gain of the base station that provides the maximum power. Superimposed on this heatmap, we plot the path taken by the drone using the RL policy when trained by using our RL algorithm (black dashed). We also superimpose, for comparison, the straight-line (i.e., shortest distance) path with a solid blue line. Fig. 3b illustrates the CDF of the SINR for each of the routes. A close examination of the RL path clearly shows that the algorithm strives to keep the drones in regions of good connectivity. For this particular case, the RL path achieves 0.2 dB better average SINR than the straight path. While this might not be significant, the true power of the RL algorithm comes to light in a dynamic environment. To illustrate this, we use the model trained on the heatmap in Fig. 3a, but evaluate it in a scenario where the center base station is removed, thereby creating a region of low coverage

in the center of the heatmap (shown in the heatmap in Fig. 3c). It can clearly be observed from the routes plotted in Fig. 3c that the straight path now directly flies over the region of poor coverage, whereas the RL path completely avoids the center and takes a different path. The CDF curve in Fig. 3d clearly illustrates that the RL path is significantly better than the straight path for this new environment.

A. Evaluations with Jammer

In order to illustrate how our RL algorithm can be employed to avoid jammers, we use the heatmap that the drone was originally trained with (heatmap in Fig. 3a). We now introduce three jammers (named A, B, C in Fig. 4a) directly in the optimized path originally predicted by RL algorithm and allow the drone to take its decisions as it flies in this new environment. As previously described, the jammers are modeled as highly directive antennas with main beam pointed directly at the drone. This significantly raises the interference and consequently drops the SINR. Thus, these jammers are characterized as deep blue spots in the heatmap shown in (Fig. 4a). The path without the presence of jammer is shown by dotted lines, whereas the path in the presence of jammer is shown by a solid line. Several very interesting phenomena are observed as the drone navigates this new environment. The drone initially follows exactly the same path as it would have if the jammer was not present. As the drone flies, it first comes near jammer A. As it senses the SINR dropping quickly near the jammer, it immediately takes the decision to change its direction and avoids the main beam of the jammer and only flies through its sidelobes, which will provide much lesser interference than the main beam. As it does this, it quickly encounters jammer B, and it is able to completely avoid it (see Fig. 4b). This is a consequence of both the path change due to jammer A as well as the RL algorithm itself ensuring that the drone only flies over regions of good SINR. Once the drone gets past jammer B, the drone almost immediately comes back to the originally planned path and continues its journey until it reaches jammer C, where it performs actions very similar to when jammer A was encountered, thereby avoiding the jammer main beam. It finally comes back to the originally planned path to reach the destination.

To completely illustrate the applicability of this algorithm to a dynamically changing wireless environment, we use the same trained model, but now evaluate it in an environment where the center base station is removed along with the presence of the jammer. The results are shown in Fig. 5, where the path without any jammers is denoted by dashed line, and the path in the presence of the jammer is illustrated in a solid black line. It is evident that the drone flies on a path identical to Fig. 3c until it reaches until it reaches the jammer location. On reaching the jammer, it skirts the area as highlighted previously. It must be re-emphasized here that the straight path, in this case, would fly through a poor coverage area and would thus lead to a drop in connectivity (denoted by the solid blue path).

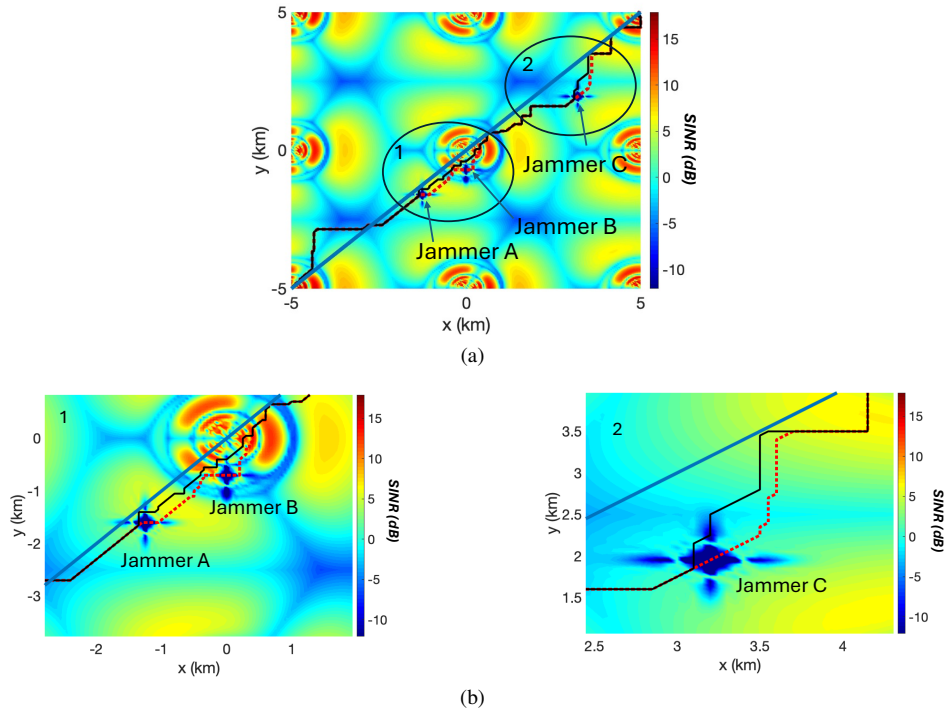


Fig. 4. (a) Comparison of RL paths obtained without jammers (red-dotted line) with the path in the presence of jammers (black solid line). The zoomed in version of this figure is shown in (b) where the deviations in the path trajectory in the vicinity of the jammer can be clearly seen. Note that there is only a slight deviation of the path near the vicinity of the jammer. Once the drone passes the jammer, the trajectory merges back with the original path.

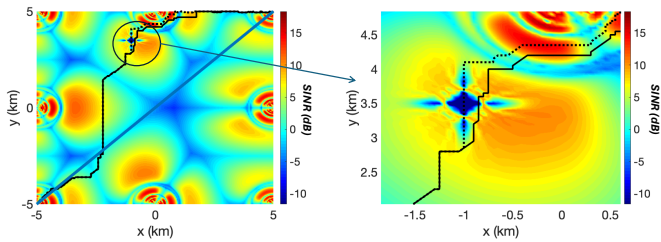


Fig. 5. Drone trajectory in an environment where there is a poor coverage zone in the center of the map with a jammer. Note that the straight path will avoid the jammer in this case, but will pass directly through a poor coverage zone. Our RL optimized path optimizes the wireless connectivity while avoiding the jammer.

These results clearly illustrate the potential of our algorithm. A comprehensive evaluation of our trained model in several other environments (for example using different jamming powers, antenna patterns, locations, etc) will be carried out in the future.

V. CONCLUSION

Drone jamming poses a serious threat to strategic missions. Concurrently, several drone applications (such as surveillance) require the drone to always fly in regions of good coverage so that data can be continuously streamed. Additionally, flying in good coverage areas is critical to maintain a stable C2 link. This paper described the development of an RL algorithm that equips the drone with the intelligence to avoid jammers

and simultaneously fly only in regions of good coverage. Specifically, we employ a constrained Q-learning approach to develop a learning table that allows the drone to take optimal actions while flying. The state space as well as the rewards were comprehensively discussed. This was followed by several representative results that clearly illustrate that our RL algorithm can concurrently optimize wireless connectivity and avoid jammers. Further, it was shown that the path deviation in the presence of a jammer is very localized leading to minimal deviations from the originally planned path. Notably, this RL model allows for the drone to be trained completely before flight. The result of the training is a look-up table that can be easily loaded into the drone, thereby adhering to the SWAP-C constraints posed by the drone.

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